

Task: Build a RAG-Based Chatbot

Objective

Create a simple RAG (Retrieval-Augmented Generation) chatbot application with a basic UI and FastAPI backend.

Requirements

Frontend

Create a simple UI where users can:

- Upload files
- Ask questions
- View chatbot responses/history

Supported File Types

- PDF (.pdf)
- DOCX (.docx)
- TXT (.txt)

Backend

Use **FastAPI** for the backend.

NOTE:

If you are shortlisted for the next round, there may be additional changes or revisions required in the current task. Make sure to keep the implementation as dynamic and flexible as possible.

Functionality

When files are uploaded:

1. Extract and process document text
2. Create embeddings and store them in a vector database
3. Implement RAG-based question answering on uploaded documents
4. Maintain chat history/context for follow-up questions

LLMs

You may use:

- Open-source/free models via:
 - Groq
 - Together AI
 - Ollama
 - Hugging Face
- Or paid APIs such as:
 - OpenAI
 - Claude

Suggested Tools

- LangChain / LlamaIndex
- FAISS / ChromaDB
- Or any other

Deliverables

- Source code with github repository
- Add README with setup instructions in github
- Working demo/UI

Evaluation Focus

- Clean implementation
- Proper RAG pipeline
- Context-aware conversation handling (User chat history maintained)
- Good backend structure UI can be as simple as you want